Practice assignment for Lesson 2.3.2

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Practice Assignment • 10 min

English

Your grade: 100%

Your latest: **100%** • Your highest: **100%**

To pass you need at least 80%. We keep your highest score.

Next item

1. What is the objective of sequential processing of measurements?

1 / 1 point

- To improve the accuracy of the Kalman-filter state estimates when z_k is a scalar.
- To improve the accuracy of the Kalman-filter state estimates when z_k is a multi-element vector.
- (selected) To speed up the execution of the Kalman filter (i.e., to reduce the number of floating-point operations required) when z_k is a multi-element vector.
- To speed up the execution of the Kalman filter (i.e., to reduce the number of floating-point operations required) when z_k is a scalar.

Correct

Yes. That is correct.

2. Which steps of the Kalman filter are modified to perform sequential processing of measurements when sensor measurements are uncorrelated? Select all that apply.

1 / 1 point

- [ ] Step 1a.
- [ ] Step 1b.
- [ ] Step 1c.
- [x] Step 2a.

Correct

Yes. This step is modified.

- [x] Step 2b.

Correct

Yes. This step is modified.

- [x] Step 2c.

Correct

Yes. This step is modified.

3. When sensor measurements are correlated, can we process the measurements sequentially?

1 / 1 point

- No, we cannot do so because the derivation assumes that $\Sigma_{\tilde{v}}$ is diagonal.
- Yes, we can do so without any modification to the process.
- (selected) Yes, but we need to modify the method by decorrelating the measurement noise via a new measurement equation $\bar{z}_k = S_v^{-1} z_k$.
- Yes, but we need to modify the method by decorrelating the process noise via a new process equation $\bar{x}_{k+1} = \bar{A}_k x_k + \bar{B}_k u_k + S_w^{-1} w_k$.

Correct

Yes. That is correct.